

Credit Card Default Auto-ML Report — ensemble (sub)

 2026-05-20 23:55:11 UTC

모형 ensemble

roc_auc = 0.7753

BEST(catboost) = 0.7796

Δ vs best = -0.0043

1 점수 요약 (본 모형 vs best)

metric	ensemble (현재)		catboost (BEST)		Δ(Test)
	Train	Test	Train	Test	
roc_auc	0.8117	0.7753	0.8099	0.7796	-0.0043
ks	0.4703	0.4236	0.4660	0.4362	-0.0126
lift	3.3114	3.0821	3.4037	3.1274	-0.0452
accuracy	0.8257	0.8168	0.8308	0.8177	-0.0008
precision	0.7103	0.6696	0.7133	0.6589	+0.0107
recall	0.3579	0.3391	0.3931	0.3640	-0.0249
f1	0.4760	0.4502	0.5069	0.4689	-0.0187
pr_auc	0.6019	0.5495	0.6194	0.5588	-0.0093

2 Feature Importance — ensemble

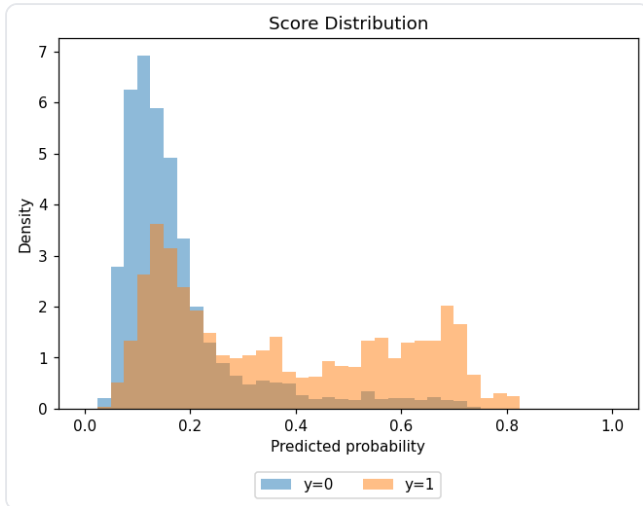
relative · top 19 · 전체 분포는 feature_importance.csv

Rank	Feature	Importance	Rank	Feature	Importance
1	PAY_0	0.4703	16	PAY_5	0.0155
2	PAY_2	0.1390	17	BILL_AMT6	0.0149
3	LIMIT_BAL	0.0435	18	PAY_6	0.0123
4	BILL_AMT1	0.0390	19	SEX	0.0028
5	PAY_AMT3	0.0352			
6	PAY_3	0.0349			
7	PAY_AMT2	0.0284			
8	PAY_AMT1	0.0256			
9	BILL_AMT2	0.0243			
10	PAY_AMT4	0.0211			
11	PAY_4	0.0210			
12	AGE	0.0196			
13	PAY_AMT6	0.0184			
14	PAY_AMT5	0.0180			
15	BILL_AMT5	0.0163			

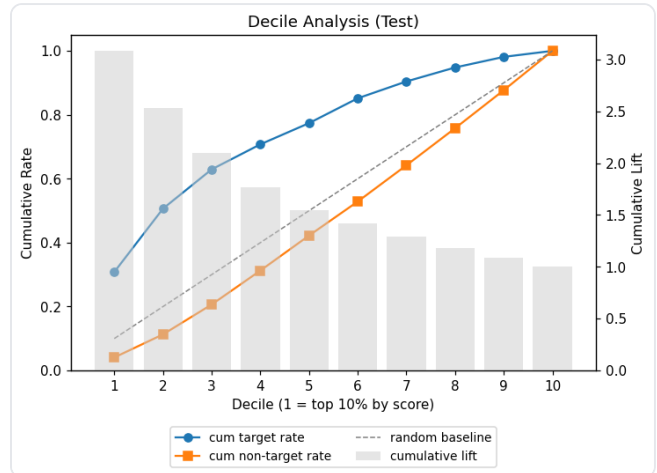
3 모델 상세 — ensemble

점수 분포 · 10-분위 차트

Score Distribution by Label



Decile Analysis Chart



Decile	Score Range	Count	Target	Non-Target	Cum Target	Cum NonTarget	Cum Count	Cum %
1	0.531 ~ 0.813	600	409	191	409	191	600	10.0%
2	0.316 ~ 0.531	600	263	337	672	528	1200	20.0%
3	0.217 ~ 0.316	600	163	437	835	965	1800	30.0%
4	0.178 ~ 0.216	600	104	496	939	1461	2400	40.0%
5	0.155 ~ 0.178	600	88	512	1027	1973	3000	50.0%
6	0.135 ~ 0.155	600	103	497	1130	2470	3600	60.0%
7	0.118 ~ 0.135	600	70	530	1200	3000	4200	70.0%
8	0.101 ~ 0.118	600	58	542	1258	3542	4800	80.0%
9	0.084 ~ 0.101	600	44	556	1302	4098	5400	90.0%
10	0.042 ~ 0.084	600	25	575	1327	4673	6000	100.0%

Confusion Matrix threshold = 0.5

	Pred 0	Pred 1
True 0	4451	222
True 1	877	450

4 변수 선택 (Stability Selection)

base: lasso

부분표본 80회

임계값 0.60

채택 19 / 23

표 노출 23 / 전체 23 (전체 분포는 메인 리포트의 feature_selection.csv)

Feature	Selection Frequency		채택
LIMIT_BAL		100.0%	✓ 채택
AGE		100.0%	✓ 채택
PAY_0		100.0%	✓ 채택
PAY_2		100.0%	✓ 채택
PAY_3		100.0%	✓ 채택
PAY_4		100.0%	✓ 채택
BILL_AMT1		100.0%	✓ 채택
PAY_AMT1		100.0%	✓ 채택
PAY_AMT2		100.0%	✓ 채택
PAY_AMT3		100.0%	✓ 채택
PAY_AMT6		100.0%	✓ 채택
PAY_5		98.8%	✓ 채택
BILL_AMT5		98.8%	✓ 채택
PAY_6		97.5%	✓ 채택
PAY_AMT4		96.2%	✓ 채택
PAY_AMT5		95.0%	✓ 채택
SEX		91.2%	✓ 채택
BILL_AMT6		88.8%	✓ 채택
BILL_AMT2		71.2%	✓ 채택
BILL_AMT3		56.2%	제외
BILL_AMT4		28.7%	제외
EDUCATION		21.2%	제외
MARRIAGE		15.0%	제외

5 학습 설정 + 튜닝 — ensemble

키	값
tuned	NO
params (최종)	{'weights': {'lgbm': 0.25285108991405325, 'xgb': 0.2538779002720508, 'catboost': 0.25449927715288256, 'elasticnet': 0.2387717326610134}}